

## INTRODUCTION TO INTERNET

Understanding computer structures can be challenging, especially when familiar terms like "**network**" and "**internet**" are used with specific technical meanings. Though they may sound similar, they represent different concepts.

A **network** is a group of two or more computers (also called hosts) connected through communication channels to share data and resources. Devices such as routers, switches, hubs, and bridges are used within a network to manage and facilitate communication between computers.

The **internet**, on the other hand, is a global network that connects millions of such individual networks across the world, enabling worldwide communication and information exchange.

### Internet Architecture

The Internet is a large network made by connecting millions of smaller computer networks all over the world. To work properly, the Internet uses a special structure called "Internet Architecture," **TCP/IP model** which is divided into layers. Each layer has a specific role and works with the layers above and below it.

The TCP/IP model is a framework that is used to model the communication in a network. It is mainly a collection of network protocols and organization of these protocols in different layers for modeling the network.

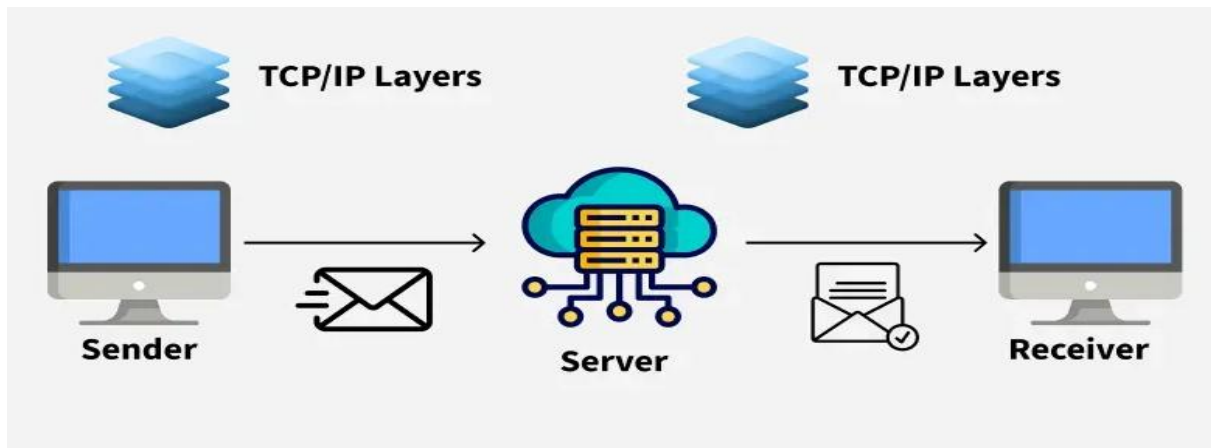
It has four layers

- Application
- Transport
- Network/Internet
- Network Access

While the OSI model has seven layers, the 4 layer TCP/IP model is simpler and commonly used in today's Internet and networking systems.

### Role of TCP/IP

One of its main goals is to make sure that the data sent by the sender arrives safely and correctly at the receiver's end. To do this, the data is broken down into smaller parts called packets before being sent. These packets travel separately and are reassembled in the correct order when they reach the destination.



## Layers of TCP/IP Model

### Application Layer

- The Application Layer is the top layer of the TCP/IP model and the one closest to the user. This is where all the apps you use like web browsers, email clients, or file sharing tools connect to the network.
- It acts like a bridge between your software (like Chrome, Gmail, or WhatsApp) and the lower layers of the network that actually send and receive data.
- It supports different protocols like HTTP (for websites), FTP (for file transfers), SMTP (for emails), and DNS (for finding website addresses).
- It also manages things like data formatting, so both sender and receiver understand the data, encryption to keep data safe, and session management to keep track of ongoing connections.

### Transport Layer

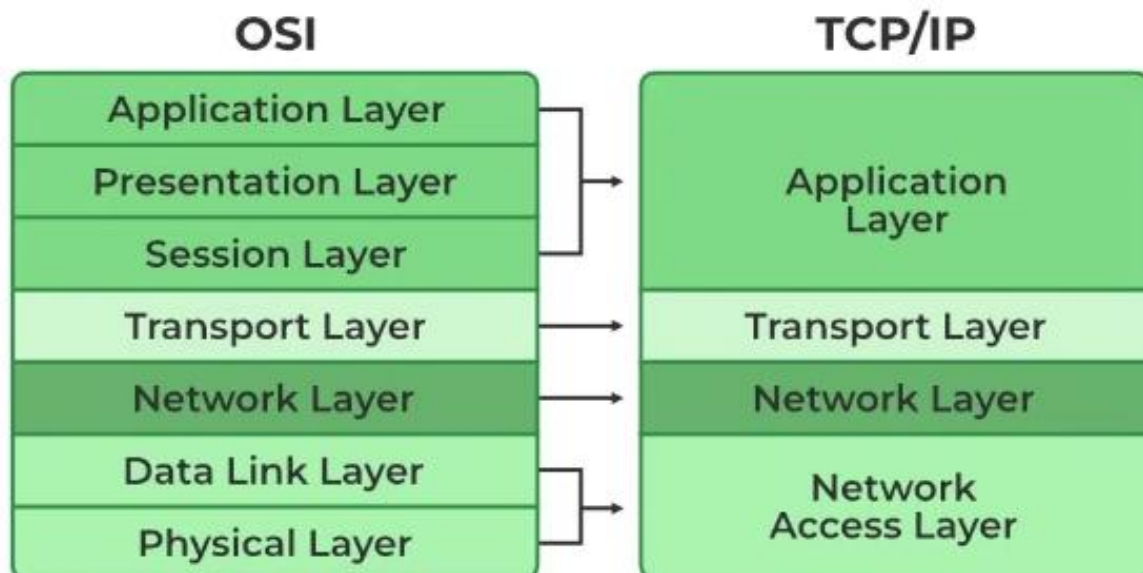
- It manages the reliable transfer of data between computers. Protocols like TCP and UDP make sure data arrives complete and in the right order.
- TCP is used when data must be correct and complete, like when loading a web page or downloading a file.
- It checks for errors, resends missing pieces, and keeps everything in order.
- UDP (User Datagram Protocol) is faster but doesn't guarantee delivery useful for things like live video or online games where speed matters more than perfect accuracy.

### Internet or Network Layer

- The Internet Layer is used for finding the best path for data to travel across different networks so it can reach the right destination. It works like a traffic controller, helping data packets move from one network to another until they reach the correct device.
- This layer uses the Internet Protocol (IP) to give every device a unique IP address, which helps identify where data should go.
- The main job of this layer is routing deciding the best way for data to travel.

### Network Access Layer

- The Network Access Layer is the bottom layer of the TCP/IP model. It deals with the actual physical connection between devices on the same local network like computers connected by cables or communicating through Wi-Fi.
- This layer makes sure that data can travel over the hardware, such as wires, switches, or wireless signals.
- It also handles important tasks like using MAC addresses to identify devices, creating frames (the format used to send data over the physical link), and checking for basic errors during transmission.



## HISTORY / EVOLUTION OF THE INTERNET

### Early Beginnings (1960s)

- The idea of networked computers started in the 1960s.
- J.C.R. Licklider, a researcher, first proposed a global network called the "Intergalactic Network."
- The U.S. Department of Defense created ARPANET in 1969, the first packet-switched network.
- Packet Switching: A method of breaking data into small packets sent to the destination independently.
- ARPANET connected four computers in different universities.

### Key Developments (1970s - 1980s)

- 1971: Ray Tomlinson sent the first email using the "@" symbol.
- The development of TCP/IP protocols standardized data transmission.
- 1983: All ARPANET computers adopted TCP/IP, the internet's foundational protocol.
- Network expansion included other academic and government systems.
- The Domain Name System (DNS) (in 1980s) was introduced, providing a hierarchical naming system for computers, services, or resources connected to the internet. This made it easier for users to navigate the growing network.

### The World Wide Web (1989-1990)

- Tim Berners-Lee invented the World Wide Web (WWW) to make information sharing easier.
- The WWW introduced:
  - Webpages
  - Hyperlinks
  - Browsers

### Internet Growth (1990s)

- The Mosaic browser (1993) made the web graphical and user-friendly.
- Commercial Internet Service Providers (ISPs) began offering access to the public.
- Popular websites like Amazon, Google, and eBay were launched.
- Domain Name System (DNS) helped translate website names into IP addresses.

**Modern Internet Era (2000s Onwards)**

- Social media platforms (Facebook, Twitter) expanded internet use.
- Mobile Internet made access easier through smartphones.
- Cloud computing enabled storing and accessing data over the Internet.
- Streaming services transformed entertainment.

**Mobile Internet and Smartphones (2007)**

The introduction of smartphones, particularly the iPhone, revolutionized internet usage by making it more accessible on mobile devices. Mobile internet usage soared, leading to the development of mobile apps and responsive web design.

**Cloud Computing and Internet of Things (IoT) (2010s)**

Cloud computing services became prevalent, allowing users to store and access data remotely. The Internet of Things (IoT) gained momentum as more devices became connected to the internet, enabling data exchange and automation.

**5G Technology and Emerging Technologies (2020s)**

The rollout of 5G technology promised even faster and more reliable internet connectivity. Emerging technologies like artificial intelligence, virtual reality, and augmented reality continued to shape and enhance online experiences.

**INTERNET FUNCTIONING**

Generally, two main components uphold the functionality of the Internet, they are:

- Packets
- Protocols

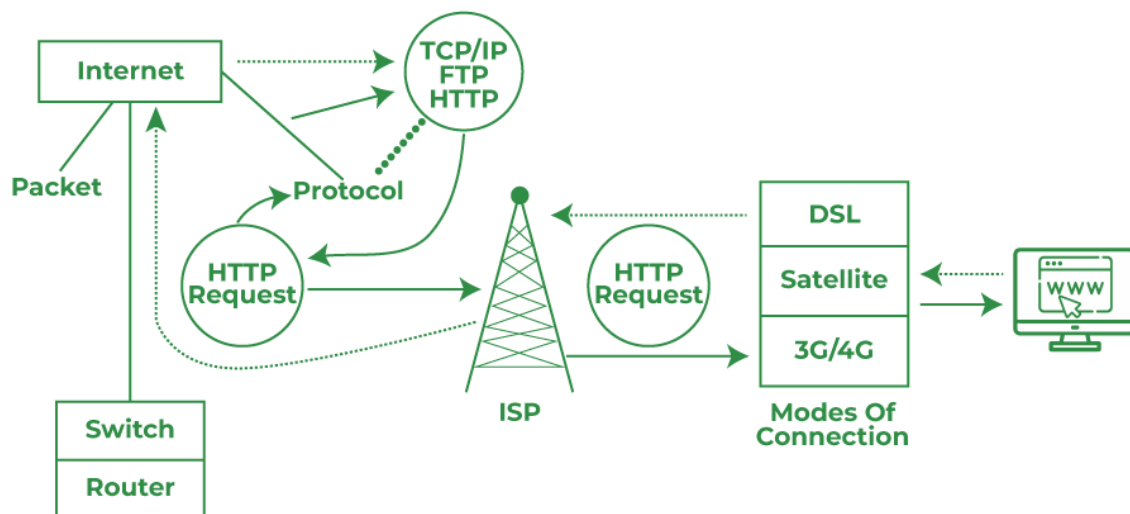
In networking, data is split into small units called packets before transmission. Each packet has part of the data plus information about its destination and order. These packets travel independently through devices like routers, taking the best available routes. Once all packets reach the receiver, they are reordered to reconstruct the original data. This process is called Packet Switching. It allows efficient and reliable transfer of large data by sending small packets.

To enable communication between computers, protocols are used. Protocols are rules that define how data is formatted, sent, and received so devices understand each other. For example, Ethernet handles data transfer within a network, IP routes packets between networks, TCP ensures reliable

delivery, HTTP formats data for websites, and UDP is used for faster transmission like video streaming or gaming.

Together, packet switching and protocols make Internet data communication flexible, efficient, and reliable.

From opening a web browser to visiting a website, it all happens with specific methods in 5 easy Steps:



1. Firstly, you'll be required to connect your system or PC with any router or modem to establish a connection. This connection is the base of the connection.
2. When you open the browser and start typing something like "www.google.com", your system will push a query command to your ISP (Internet Service Provider) that is connected with other servers that store and process data.
3. Now, the web browser will start indexing the URL that you've entered and will fetch the details in numeric format (in their own language to identify the address (unique) that you're trying to reach).
4. Next is, now your browser will start sending the HTTP request where you're trying to reach and sends a copy of the website on the user's system. Note: The server will send data in the form of small packets (from the website to the browser).
5. Once all the data (of small packets) will be received at the user's end (PC/Laptop), the browser will start arranging all those small packets and later will form a collective file (here, the browser will gather all the small packets and rearrange them just like a puzzle) and then you'll be able to see the contents of that website.

### Modes of connecting through the internet

**DSL:** This technology uses a Broadband connection which is in trend for the past few years. ISP will connect our premises with the help of telephone wire despite the fact that own any telephone.

**Dial-Up:** People used to connect their system with the help of a dial-up connection, and it is one of the slowest types of Internet connection. This is used to enable internet connectivity with the help of a telephone connection and the user must have multiple connections then only they can use a Dial-up connection.

**Cable TV Connection:** It is being used to connect our system with the Internet, and for that, ISP will connect it via cable TV wire. It also uses Broadband technology and really don't need to have a Cable connection for that. Cable is being considered as most accessible as and faster than dial up and DSL that we have for connection.

**Satellite:** It also uses broadband technology but without interacting with any cable connection. Hence, it connects wirelessly with the help of a satellite and this enables its availability anywhere in the world. Thus, being fancy and accessible it comes with a few drawbacks:

- a) There can be network disturbance if the weather is unstable because it connects via satellite.
- b) The connectivity is not stable and they are considered slower as compared to DSL or cable connection.

**3G/4G/5G:** This is the new age technology in the entire world. It connects wirelessly via different ISPs and is widely used in cellphones. But they aren't being considered as stable that DSL or cable and most importantly they come with a DATA LIMITATION cap for each month.

### Internet Connection Protocols

The protocols decide how the technology is going to work and in what governing ways and what would be the quantity of data will be shared. They all are defined by protocols and as per standard, both the parties (sender/receiver) have to follow the same rule in order to communicate. However, these protocols are categorized into 3 major units:

1. **TCP/IP:** It is a variety of internet protocols for communicating between a sender and a receiver. TCP (Transmission Control Protocol) / IP (Internet Protocol) ensures that all users who are connected to the internet have their own unique identity known as IP Address. However, how data will flow into what segments (packets) is being decided by TCP. IT

divides the whole message into small packets and reassembles them before it reaches the receiver.

2. **FTP:** It is used for communicating from one point to another (computer) over LAN(Local Area Network) or WAN (Wide Area Network). File Transfer Protocol or FTP acts like a host and establishing connections between computers and transferring files between them are seamless whereas one connection is used for data connection and the other for the control connection.
3. **HTTP:** The base of the Internet starts from HTTP (Hypertext Transfer Protocol) and is being used to communicate all over (World Wide Web) WWW. The concept of HTTP is to provide data over the Internet (image, video, text, etc.) and the moment a user opens their web browser, they're connecting to the internet and establishing an HTTP connection. The foundation of HTTP began in the '90s by Tim Berners-Lee which runs on top of the TCP/IP network.

### Pillars of the Internet

**Switches:** Switches help for connecting devices in a single network and use small packet switching to send and receive data packets over the Internet. Besides this, they have multiple ports by which a system can be connected. So, when a packet arrives at any port, it starts cross-checking the specifications and forward the same to its destination. It also supports broadcast and unicast communication.

**Router:** They operate at the layer 3 OSI (Open Systems Interconnection Model) model that is created for sending, receiving, and forwarding small data packets within the connected system over the same network. In this architecture, once the router receives the data packets, it inspects the destined address, then consults its routing and transfers the same packet to its desired location.

### Main applications of the Internet

- **Email:** Sending and receiving messages worldwide.
- **Web Browsing:** Visiting websites to get information, news, and entertainment.
- **E-commerce:** Buying and selling products online (e.g., Amazon, Flipkart).
- **Online Banking:** Managing accounts and transferring money electronically.
- **Social Media:** Connecting with friends and sharing content (e.g., Facebook, Instagram).
- **Streaming:** Watching videos, listening to music (e.g., YouTube, Spotify).
- **Online Education:** Learning through platforms like Coursera, Khan Academy.
- **Remote Work:** Collaborating on projects using tools like Google Drive, Zoom.
- **Telemedicine:** Consulting doctors online.

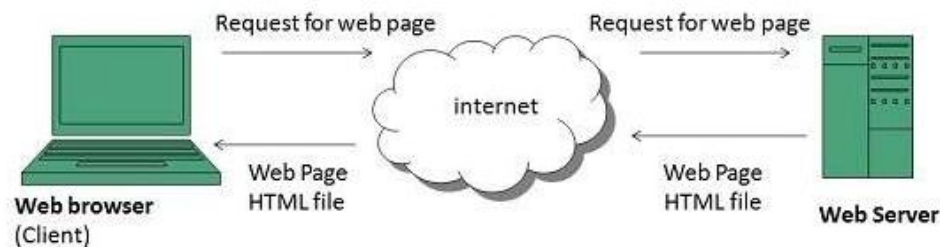


- **Travel & Navigation:** Planning trips and navigation via Google Maps.
- **Digital Payments:** Making instant payments using apps like Paytm, Google Pay.

## INTERNET SERVICES

### WORLD WIDE WEB (WWW)

The World Wide Web (WWW or Web) is a major Internet tool for publishing and accessing information. It was created by Tim Berners-Lee in 1989 at CERN, Switzerland. The Web is a hypertext-based, multimedia, and distributed information system. It connects documents and resources through links and supports various Internet tools. Its unique features make it the most advanced and widely used system on the Internet.

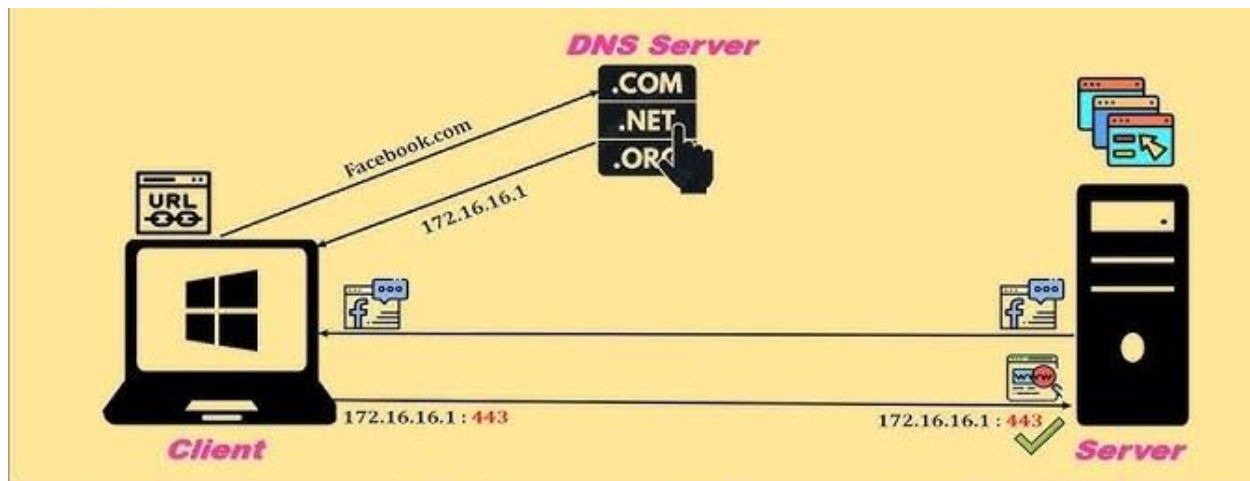


### *Importance of the World Wide Web (WWW)*

The World Wide Web (WWW) is a system that connects billions of people worldwide, making it easy to access and share information from anywhere.

### **Key points on the importance of the WWW:**

- The Web allows fast and easy access to a vast range of information, supporting education, research, and learning.
- It enables communication and collaboration globally, helping people work and share ideas without geographic boundaries.
- Businesses use the Web to reach customers worldwide, conduct online trade, and provide services efficiently.
- The Web empowers individuals to publish content, share knowledge, and participate in social and professional networks.
- Overall, the WWW has transformed how society communicates, learns, works, and entertains, making it a foundation of the modern digital world.

**Working of www:**

The World Wide Web (WWW) is a system that allows people to access and share information over the Internet using web browsers. Following is a step-by-step explanation of how the Web works:

1. **Web Browser and URL:** You start by typing a web address, called a URL (Uniform Resource Locator), into a web browser (like Chrome or Firefox). This URL tells the browser which web page you want to see.
2. **Domain Name System (DNS):** The browser sends the website name (like `www.example.com`) to a DNS server. The DNS translates this into an IP address, which is like the website's phone number on the Internet.
3. **Request to Web Server:** The browser then sends a request to the web server at that IP address, asking for the web page files. This request uses a protocol called HTTP (Hypertext Transfer Protocol) or HTTPS for secure connections.
4. **Web Server Response:** The web server receives the request and sends back the web page, usually written in HTML (Hypertext Markup Language). This page can include text, images, videos, and links.
5. **Rendering Web Page:** Your browser reads the HTML and other files (like CSS for styles and JavaScript for interaction) and displays the page visually on your screen.
6. **Navigation via Hyperlinks:** Web pages contain hyperlinks (clickable links) that let you easily jump from one page to another, creating a network of connected information.

This entire process is based on a **client-server model** where the browser is the client requesting data and the web server provides it. HTTP/HTTPS protocols and technologies like HTML make

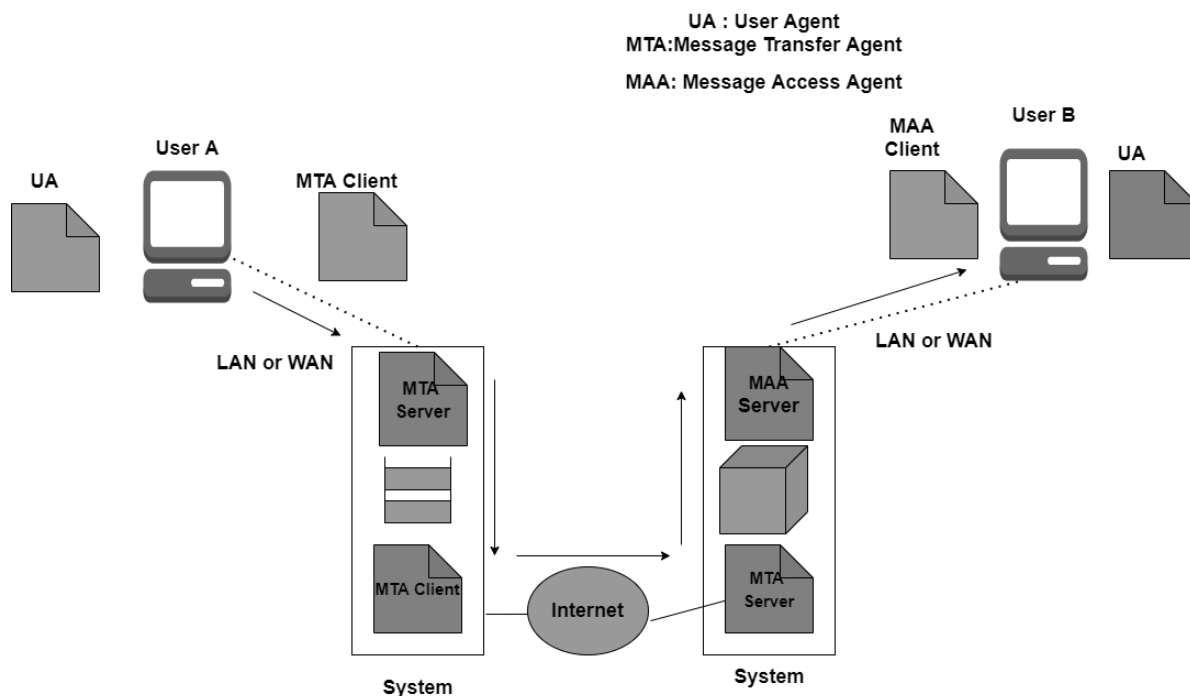
the web fast, flexible, and easy to use. This system supports multimedia content and interactive services, making the WWW a powerful tool for education, business, and daily communication.

## ELECTRONIC MAIL (E-mail)

Electronic Mail (E-mail) is a fast, easy, and cost-effective method of communication over the Internet. It is the most widely used Internet service, allowing users to send and receive messages instantly across the world. Apart from personal communication, e-mail is used to share documents, exchange research data, subscribe to online journals or news services, and access digital libraries. Users can also attach various file types such as text, images, and spreadsheets to their messages, making e-mail a versatile communication tool.

E-mail is a fast and convenient method of communication. Messages travel through high-speed networks and usually reach their destination within minutes, making traditional postal mail seem very slow, often called “snail mail.” Unlike paper letters, e-mails can be stored, edited, and printed easily using a computer. They can also be forwarded or sent to multiple recipients instantly, eliminating the need for envelopes, stamps, and physical delivery.

The person who is sending mail is called **sender** and person who receives mail is called **recipient**.



Key protocols like SMTP (**Simple Mail Transfer Protocol**) used for sending and POP3 (**Post Office Protocol 3**) or IMAP (**Internet Message Access Protocol**) for receiving manage this process.

The basic components of an email system are : User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. These are explained as following below.

**User Agent (UA) :** The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.

**Message Transfer Agent (MTA):** MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfers mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple Mail Transfer Protocol.

**Mailbox:** It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement. To use e-mail system each user must have a mailbox. Access to mailbox is only to owner of mailbox.

**Spool file:** This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery.

**Mail Access Agent (MAA):** MAA is a part of the email system that retrieves incoming messages from a mail server to a user's computer, using protocols like POP3 or IMAP. While the Message Transfer Agent (MTA) handles sending emails between servers, the MAA is the component the end-user connects to for accessing and downloading their emails from the server's mailbox.

### Advantages

1. Convenient and fast communication with individuals or groups globally.
2. Easy to store and search for past messages.
3. Ability to send and receive attachments such as documents, images, and videos.
4. Cost-effective compared to traditional mail and fax.
5. Available 24/7.

### Disadvantages

1. Risk of spam and phishing attacks.
2. Overwhelming amount of emails can lead to information overload.
3. Can lead to decreased face-to-face communication and loss of personal touch.
4. Technical issues, such as server outages, can disrupt email service.
5. It is important to use email responsibly and effectively, for example, by keeping the subject line clear and concise, using proper etiquette, and protecting against security threats.

### FILE TRANSFER (FTP)

File transfer is one of the most common Internet applications used to copy files from various online archives. These archives act as electronic libraries that store digital information such as text, images, audio, video, and software programs.

**File Transfer Protocol (FTP)** is the standard Internet protocol for transferring files between computers. It follows the client–server model, where the FTP client connects to an FTP server to upload or download files.

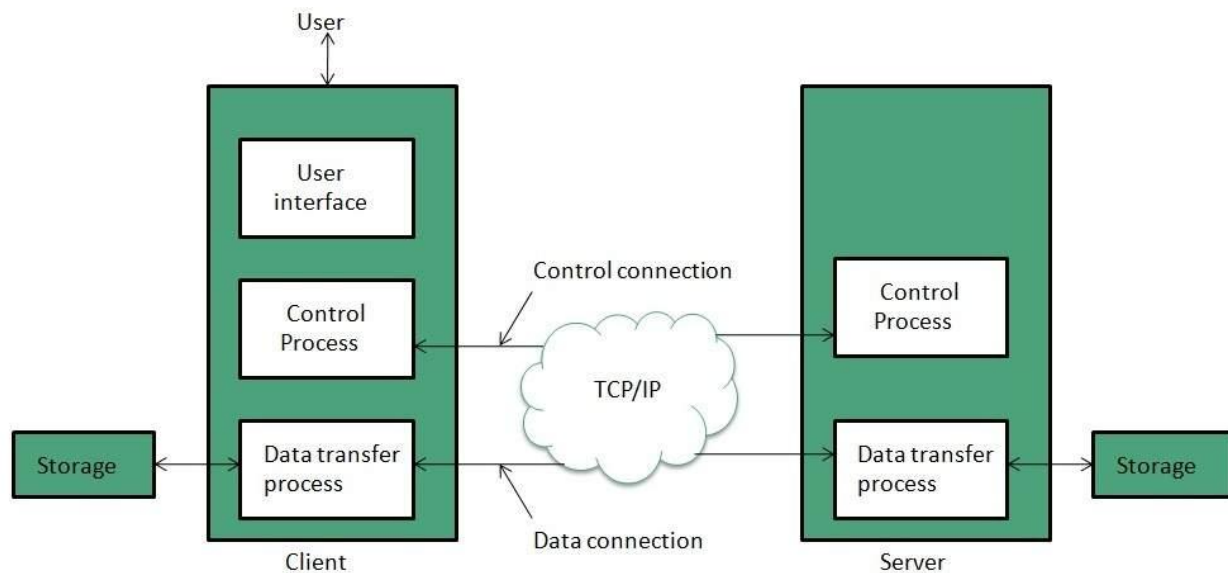
### Types of FTP Connections

- **Anonymous FTP:** Allows public access without login credentials. Users log in with the username “anonymous” and their email address as the password.
- **Non-Anonymous FTP:** Requires a valid username and password for secure access to private files.

### Modes of File Transfer

- **ASCII Mode:** Used for transferring text files.
- **Binary Mode:** Used for all other types of files, such as images, software, and multimedia. (When unsure, use binary mode.)

Following diagram illustrates how the File Transfer Protocol (FTP) works to transfer files between a client and a server over the Internet using TCP/IP.



## Key Components

### ➤ Client Side:

The user interacts with the FTP client through a user interface. The client uses a control process to manage the connection setup and command execution, and a data transfer process to handle the movement of files to or from storage.

### ➤ Server Side:

The server has its own control process that manages the commands received from the client. Its data transfer process handles transferring files to or from its own storage system.

## Process Explanation

### ➤ Control Connection:

The client establishes a control connection with the server, which is used to send commands (such as login credentials, navigate directories, or request file operations) between the client and server control processes.

### ➤ Data Connection:

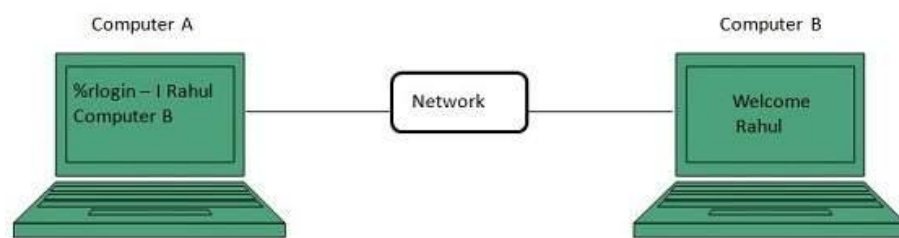
When an actual file transfer is requested, a separate data connection is established through TCP/IP between the client and the server. The data transfer processes on both sides manage the exchange of file data between their respective storage systems.

➤ **Storage Access:**

The client and server each have access to their own storage systems, allowing files to be uploaded or downloaded as needed through the data transfer process.

## TELNET

Telnet is a network protocol that allows users to remotely log in and control another computer over a network, typically the Internet. This protocol provides basic text-based access to a remote system, as if the user were physically present at its terminal.



➤ **Client-Server Model:**

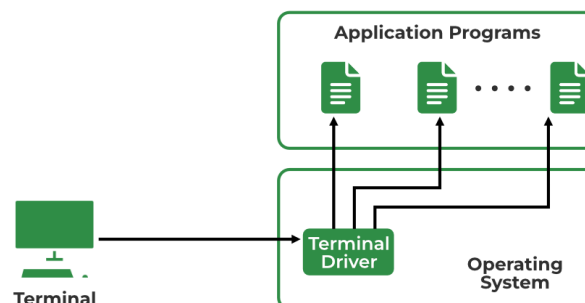
Telnet operates on a client-server model. The user starts a Telnet client program on their local computer (Computer A). Using Telnet commands, the user can connect to a remote computer (Computer B) over the network.

## Logging in TELNET

The logging process can be further categorized into two parts:

- Local Login
- Remote Login

**Local Login:** Whenever a user logs into its local system, it is known as local login

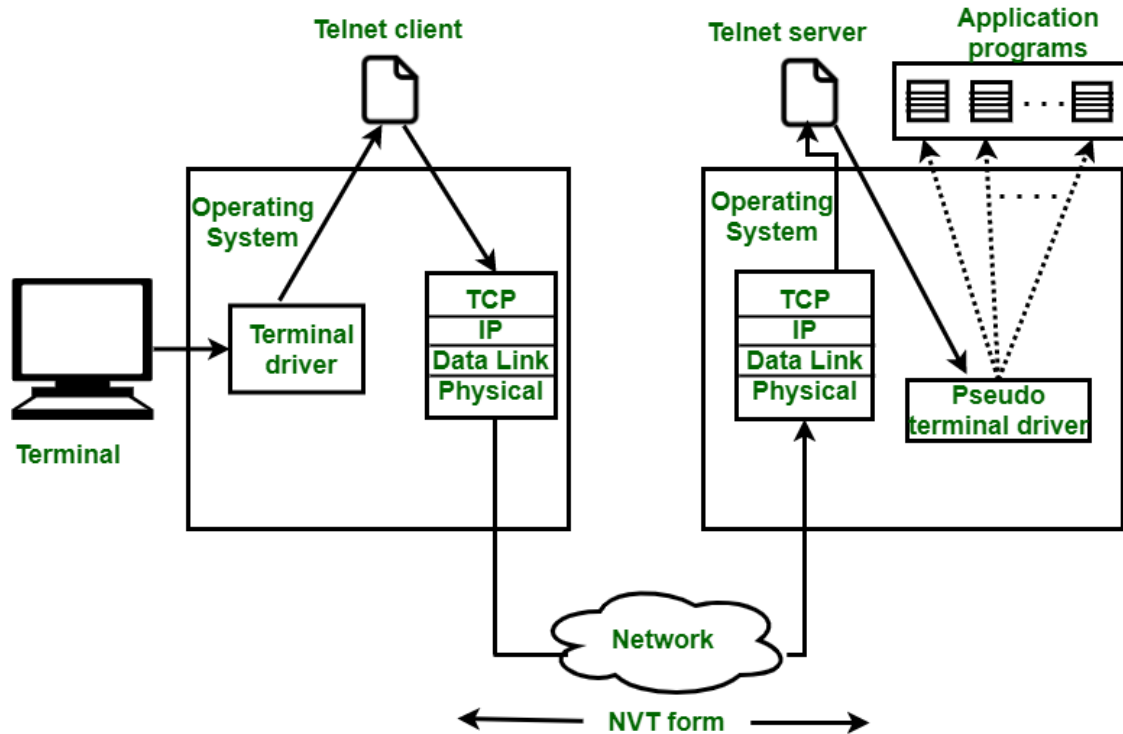


### The Procedure of Local Login

- Keystrokes are accepted by the terminal driver when the user types at the terminal.
- Terminal Driver passes these characters to OS.
- Now, OS validates the combination of characters and opens the required application.

### Remote Login

Remote Login is a process in which users can log in to a remote site i.e. computer and use services that are available on the remote computer. With the help of remote login, a user is able to understand the result of transferring the result of processing from the remote computer to the local computer.



### The Procedure of Remote Login

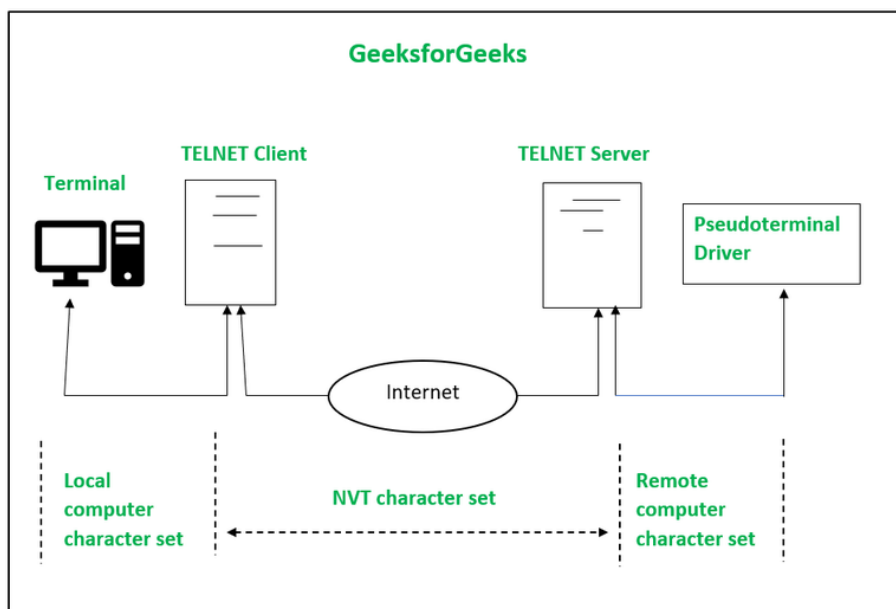
- When the user types something on the local computer, the local operating system accepts the character.
- The local computer does not interpret the characters, it will send them to the TELNET client.
- TELNET client transforms these characters to a universal character set called Network Virtual Terminal (NVT) characters and it will pass them to the local TCP/IP protocol Stack.



- Commands or text which is in the form of NVT, travel through the Internet and it will arrive at the TCP/IP stack at the remote computer.
- Characters are then delivered to the operating system and later on passed to the TELNET server.
- Then TELNET server changes those characters to characters that can be understandable by a remote computer.
- The remote operating system receives characters from a pseudo-terminal driver, which is a piece of software that pretends that characters are coming from a terminal.
- The operating system then passes the character to the appropriate application program.

### Network Virtual Terminal (NVT)

NVT (Network Virtual Terminal) is a virtual terminal in TELNET that has a fundamental structure that is shared by many different types of real terminals. NVT (Network Virtual Terminal) was created to make communication viable between different types of terminals with different operating systems.



### TELNET Working

- **Client-Server Interaction**
  - The **Telnet client** initiates the connection by sending requests to the Telnet server.
  - Once the connection is established, the client can send **commands** to the server.
  - The server processes these commands and responds accordingly.
- **Character Flow**

- When the user types on the **local computer**, the local operating system accepts the characters.
- The Telnet client transforms these characters into a universal character set called **Network Virtual Terminal (NVT)** characters.
- These NVT characters travel through the Internet to the remote computer via the local TCP/IP protocol stack.
- The remote Telnet server converts these characters into a format understandable by the remote computer.
- The remote operating system receives the characters from a pseudo-terminal driver and passes them to the appropriate application program.
- **Network Virtual Terminal (NVT)**
  - NVT is a virtual terminal in Telnet that provides a common structure shared by different types of real terminals.
  - It ensures communication compatibility between various terminals with different operating systems.

➤ **User Experience:**

After logging in through Telnet, the user on Computer A can perform operations on Computer B as if directly interacting with its console. This includes running commands, managing files, or administering the system—all via a simple command-line interface.

## Features and Uses

➤ **User-Friendly Interfaces:**

While Telnet was originally text-based, modern Telnet clients sometimes provide graphical user interfaces to simplify remote access for users, making it easier to connect and interact with remote systems.

➤ **Applications:**

Telnet is especially useful for remote administration, troubleshooting, software development, and network testing. However, because it does not encrypt data (sending user credentials and commands as plain text), it is being replaced by more secure protocols like SSH (Secure Shell) for sensitive operations.

Telnet's lack of encryption, means it should not be used for transferring confidential information or on unsecured networks. For secure remote login, SSH is preferred in modern environments.

## GOPHER

Gopher is an early Internet protocol designed to make finding, retrieving, and exploring information on remote computers fast and simple, especially before the World Wide Web became popular.

**Definition:** Gopher was developed in 1991 at the University of Minnesota as a client-server system that uses a menu-driven, hierarchical structure to organize and access text documents and files over the Internet. It provided an intuitive way for users to browse directories and select files, similar to working with folders and files on a local computer.

### How Gopher Works

- **Client-Server Model:**  
Users interact with a Gopher client application on their computer. This client connects to a Gopher server, typically over TCP port 70.
- **Browsing Menus:**  
The server sends a list of menus or files. Users browse these menus, which might point to documents, directories, or even links to other Gopher servers.
- **Retrieving Files:**  
When a user selects a file or document, the client requests it from the server, and the file is transferred and displayed on the local system.

### Key Features of Gopher

- **Hierarchical Navigation:**  
Information is organized into menus and submenus, resembling a large library or file system, helping users find content efficiently.
- **Text-Based Interface:**  
Gopher is lightweight and focuses on quick access to text files, making it fast even on slow connections and easy for beginners to use.
- **Extensible:**  
Although aimed at text files, Gopher can also support other types of files and connect to various resources.

### Practical Use and Benefits

- **Efficiency:** Gopher transfers require low bandwidth, making it suitable for slow or limited connections.
- **Content Focus:** Its simple design emphasizes information, not graphics or multimedia.
- **Educational and Retro Computing:** Some academic institutions and hobbyists still use Gopher for sharing research papers or as part of retro computing communities.

### Qns: Why Gopher Became Less Popular?

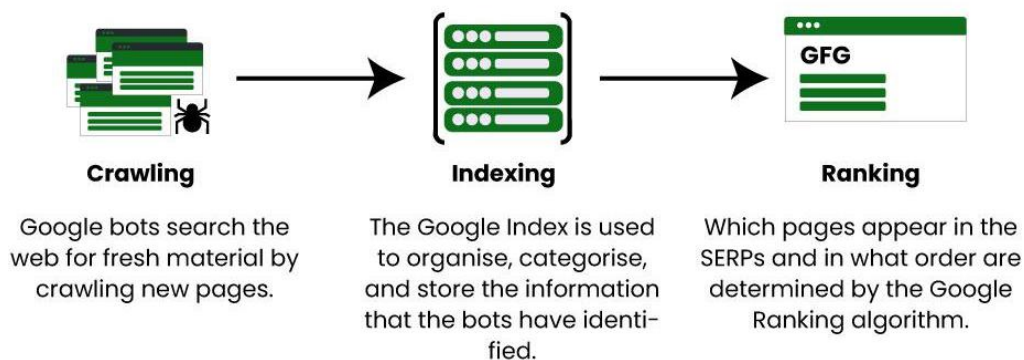
**Ans:** Gopher became less popular because licensing fees were introduced, the World Wide Web (www) offered more flexible navigation and better multimedia support, and the Internet community shifted its focus to web technologies that were free and more advanced.

## SEARCH ENGINE

A search engine is one of the most essential basic services of the Internet, helping users locate information quickly and efficiently from the vast amount of content available online.

**Definition:** A search engine is a specialized software system designed to search for information on the World Wide Web. It allows users to enter keywords or questions, and then displays a list of websites, documents, images, or other resources related to the search query.

### How Search Engines Work



- **Crawling:** Search engines use automated programs called "crawlers" or "spiders" or "bots" to browse and scan web pages across the Internet. These crawlers continuously discover and update information. The information which is gathered by web crawler by crawling through internet is stored on the database.

- **Indexing:** The collected information is organized in a massive database, called an index. This index helps the search engine quickly retrieve relevant data when a user makes a query. Information identified by the crawler needs to be organized, Sorted, and Stored so that it can be processed later by the ranking algorithm.
- **Ranking:** Ranking is the position by which your website is listed in any Search Engine. There is following three steps in which how ranking works.
  - ✓ **Query Processing:** When a user enters keywords or questions, the search engine compares them against its indexed data. It then sorts and ranks the results based on relevance, quality, and other factors.
  - ✓ **Finding matching pages** - This step is to look into their index and find the best matching pages, for example, if you search dark wallpaper then it gives you the result of images, not text.
  - ✓ **Present the results to the users** - A typical search results page includes ten organic results in most cases it is enriched with other elements like paid Ads, direct answers for specific queries, etc.

### *Examples of Popular Search Engines*

- Google
- Bing
- Yahoo!
- DuckDuckGo

### *Importance of Search Engines*

- **Easy Access to Information:** Search engines make it easy to find articles, videos, research papers, images, and other content on virtually any topic.
- **Educational Utility:** They are valuable tools for students, researchers, and professionals to find specific information and stay updated.
- **Navigation and Discovery:** Search engines help users discover new websites and sources by suggesting relevant links.

## NEWS GROUP

A newsgroup is an online discussion forum or a collection of messages on the Internet where people post and read about specific topics. It works as a place to share information, ask questions, and discuss ideas with others interested in the same subject. Newsgroups are part of the Usenet system, which was created before the World Wide Web and is organized into different categories based on topics.

### *How Newsgroups Work*

- Newsgroups use a special protocol called **Network News Transfer Protocol (NNTP)** to send and receive messages between servers and users.
- Users access newsgroups through **newsreader software**, which is similar to an email program but designed to read and post messages to groups rather than individuals.
- Messages posted by users are called **posts** or **articles** and are stored on news servers that replicate the posts to other servers, making discussions available worldwide.
- Newsgroups are organized by names that describe their topics, such as comp.software (for computer software) or sci.physics (for science and physics discussions).
- There are thousands of newsgroups covering a wide range of topics including technology, hobbies, politics, and more.

### *Types of Newsgroups*

- **Moderated Newsgroups:** Posts must be approved by a moderator before they appear to others. This keeps discussions on topic and reduces spam.
- **Unmoderated Newsgroups:** Posts appear immediately without approval, allowing faster communication but with less control over content.
- **Text Newsgroups:** Mainly for discussions and sharing text-based information.
- **Binary Newsgroups:** Users share files like images, videos, or software, often split into parts for easier download.

### *Uses and Importance*

- Newsgroups were one of the earliest forms of online communities for sharing knowledge and solving problems collaboratively.
- They helped in software development, academic discussions, hobbyist groups, and special interest communities.
- Though modern forums and social media have largely replaced them, newsgroups are still used for specialized discussions and as a file-sharing platform.

### *Advantages of Newsgroups*

- Provide a platform for wide-ranging public discussions.
- Enable asynchronous communication allowing users to participate anytime.
- Organize information into well-defined topic groups.
- Distributed nature means data is stored across many servers making it robust and reliable.

## **CHAT ROOMS**

Chat rooms are a basic service of the Internet that allows users to communicate with others in real time through text messages in a shared online space.

**Definition:** A chat room is a virtual platform hosted on a server where multiple users can join and exchange messages instantly. Conversations take place inside a visible chat box, and messages typed by any participant are seen by all other members in that room. Chat rooms may be public or private, and are often organized by topic or interest.

### **Key Features**

- **Real-time communication:** Users interact simultaneously, with messages appearing immediately for everyone in the chat room.
- **Group interaction:** Unlike instant messaging (which is typically one-to-one), chat rooms allow group discussions with many users at once.
- **User accounts:** Many platforms require the creation of an account to join and participate in chat rooms.
- **Customization:** Modern chat rooms often feature private messaging, topic-specific rooms, and customization options.

**Usages:** Chat rooms are used for various purposes including casual conversations, professional collaborations, gaming communities, social networking, and interest-based discussions.

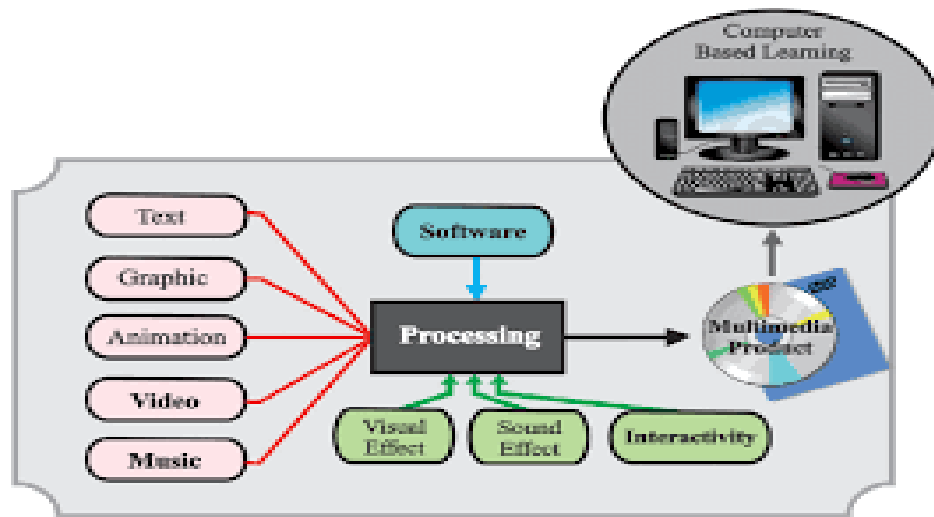
**Examples:** Popular platforms offering chat room functionality include Discord, Slack, Microsoft Teams, and traditional services like IRC (Internet Relay Chat).

## **MULTIMEDIA**

The word **multi** and **media** are combined to form the word multimedia. The word "multi" signifies "many." the means of **media** communication that reach or influence people widely.

Multimedia refers to the integration of multiple forms of media, such as text, images, audio, video, and animations, in a digital format to create engaging and interactive content. It enhances data presentation and user experience by combining sensory elements, making information more accessible, memorable, and impactful.

**Definition:** Multimedia is a combination of different media types used to communicate information effectively through digital technology.



### Characteristics:

- **Interactivity:** Users can actively engage with multimedia content.
- **Rich content:** Combines visuals, sounds, and motion for comprehensive understanding.
- **Multisensory stimulation:** Uses sight, sound, and sometimes touch to create immersive experiences.
- **Digital format:** Compatible with computers and the internet, allowing easy editing, sharing, and storage.

### Components of Multimedia

#### ➤ Text

- Fundamental for conveying detailed information.
- Utilized in titles, captions, descriptions, and hypertext links.
- Formats: Plain text, HTML, rich text formats.

#### ➤ Graphics

- Static images, diagrams, charts, icons, and illustrations.
- **Formats:** JPEG, PNG, GIF, SVG for scalable vector graphics.
- **Purpose:** Visual representation to support or clarify textual content.



➤ **Audio**

- Sound recordings, narration, music, sound effects used to enhance emotional and informational impact.
- **Formats:** MP3, WAV, AAC.
- **Application:** Voice-overs for tutorials, background music in videos, audiobooks.

➤ **Video**

- Combines moving images with sound for storytelling, education, or advertising.
- **Formats:** MP4, AVI, MOV.
- Enhances engagement by providing real-world visual context.

➤ **Animation**

- Sequence of static images creating the illusion of motion.
- **Types:** 2D animation (cartoons), 3D animation (modeling).
- **Tools:** Adobe Animate, Blender.

➤ **Interactivity**

- User engagement through clickable links, quizzes, simulations, games, etc.
- Essential for e-learning and digital training modules.

### Multimedia Software Tools

- **Text Editors:** Used to create and edit text content (MS Word, Notepad).
- **Image Editing Software:** Adobe Photoshop for editing graphics, CorelDRAW for designing vector graphics.
- **Audio Software:** Audacity lets users record and edit audio clips, useful for podcasts and presentations.
- **Video Editing Tools:** Adobe Premiere and Filmora help create and edit professional videos.
- **Animation Tools:** Adobe Animate and Blender create 2D and 3D animations.
- **Authoring Tools:** Software like Adobe Captivate creates interactive educational content integrating all multimedia elements.

### Applications of Multimedia

- **Education:** Online courses, virtual labs, language learning apps. Multimedia helps clarify concepts with visual and audio aids.
- **Entertainment:** Movies, music videos, animated cartoons, and video games rely heavily on multimedia.
- **Advertising:** Interactive ads use multimedia to engage customers more effectively.
- **Communication:** Video conferencing apps use audio, video, and text together.
- **Training and Simulation:** Used in medical, military, and aviation fields to create realistic practice scenarios without real-world risks.

### TRANSITION FROM CONVENTIONAL MEDIA TO DIGITAL MEDIA

The transition from conventional media to digital media represents one of the most significant shifts in the communication and information landscape over recent decades. Conventional media, such as newspapers, radio, and television, traditionally operated on a one-way communication model where content was produced and controlled by a few organizations and disseminated to a passive audience. Digital media, by contrast, leverages internet technologies and digital devices to enable a highly interactive and instantaneous flow of information accessible anytime and anywhere.

### Characteristics of Conventional Media

- **One-way communication:** Content flows from media producers to audiences without direct feedback.
- **Fixed schedules and formats:** Programming and publications follow planned and rigid schedules.
- **High cost and gatekeepers:** Production, distribution, and publishing require significant investment and control by established entities.
- **Limited reach:** Traditional media mostly serves regional or national audiences.
- **Examples:** Print newspapers, broadcast TV and radio, magazines.

### Features of Digital Media

- **Interactive communication:** Users can engage, comment, share, and create content themselves.
- **On-demand access:** Content is available whenever and wherever the audience chooses.
- **Low barriers to entry:** Anyone with internet access can publish and distribute content.
- **Global reach:** Digital content transcends geographical boundaries instantly.
- **Multimedia integration:** Combines text, audio, video, animation, and interactive elements seamlessly.

- **Examples:** Social media, blogs, streaming platforms, podcasts, online news sites.

### Democratization of Content Creation and Distribution

One of the most profound impacts of digital media is the democratization of content creation. Previously, content creation was restricted to professionals with access to expensive equipment and distribution networks. Now, affordable smart phones, cameras, and user-friendly software tools empower individuals and small organizations worldwide to create, publish, and monetize their content. Platforms like YouTube, Instagram, TikTok, and blogging sites have dramatically expanded the pool of content creators, allowing diverse voices to be heard.

### Impact on Traditional Media

In response to digital transformation, many traditional media outlets have:

- Developed their digital platforms including websites, apps, and social media presence.
- Adopted multimedia content, such as incorporating video articles and podcasts.
- Faced financial challenges due to shrinking advertising revenues as marketers shift digital.
- Encountered competition with online influencers and independent content creators.
- Struggled with misinformation and decreasing trust, as digital platforms sometimes lack editorial rigor.

### Challenges in the Digital Age

- **Misinformation and quality control:** Digital platforms facilitate rapid spread of unverified or false information.
- **Monetization:** Creators must find sustainable revenue models amid changing viewer behavior.
- **Content saturation:** Enormous volumes of content make discoverability and audience engagement challenging.
- **Digital divide:** Access to technology and skills remain uneven globally.